BIOCH 590 **Biomacromolecules** **Spring 2009**

Part I: January 16 – March 6

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Office Hours: by appointment

Recommended Textbooks (Available from Hale Library Reserves):

The lecture notes can be used as the main reference for study, while the following textbooks will provide a more comprehensive source of information on the related topics.

1. *Principles of Physical Biochemistry* by van Holde, Johnson, and Ho (1998 or newer)

2. *Physical Chemistry: Principles and Applications in Biological Sciences* by Tinoco, Sauer, Wang and Puglisi (4th Edition)

Tentative Topics

1. Overview of thermodynamics: first and second laws, free energy, chemical equilibrium
   Main References: Lecture notes, Chap 2 of van Holde

2. Introduction to statistical mechanics: Boltzmann distribution, partition function, connection to thermodynamics
   Main References: Lecture notes, Chap 11 of Tinoco (pages 645-658)

3. Molecular interactions and biomolecular structures: types interactions, stabilizing forces, helix-coil transitions, calorimetry
   Main References: Lecture notes, Chap 2-3 of van Holde, Chap 9 of Tinoco

4. Molecular motion and transport: diffusion, viscosity, electrophoriesis
   Main References: Lecture notes, Chap 5 of van Holde

5. Computer modeling of biomolecules: molecular mechanics, force fields, simulation techniques (energy minimization, Monte Carlo, molecular dynamics)
   Main References: Lecture notes

6. Macromolecular interactions: protein-ligand, protein-protein and protein-nuclei acid inteactions, cooperativity, allosteric effects
   Main References: Lecture notes, Chap13-14 of van Holde

Exams: **Hour Exam 1: Feb. 9, Monday** **Hour Exam 2: Mar. 6, Friday**

Grading: The grade for Part I will be assigned based on both exam scores (70%) and additional homework assignments (30%). Parts I & II will contribute equally to the final grade.

Notes

1. Any student with a disability requiring special assistance in this course should contact Disability Support Services and speak with the instructors as soon as possible.
2. Please note the following statement of policy by the Faculty Senate: "*Plagiarism and cheating are serious offenses and may be punished by failure on the exam, paper or project; failure in the course; and/or expulsion from the university*." For more information, visit http://www.ksu.edu/honor.